

Conflict, Persistence, and Route Switching: Migration Corridor Allocation Under Changing Transit Risk

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April 2026

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1 Introduction

The allocation of irregular migration flows across transit corridors is a central concern for both migration policy and the welfare of migrants in transit. When conflict escalates along a migration route or at the origin, migrants face a choice: persist on established corridors or redirect through alternative routes. Understanding the determinants of this allocation, and the degree of persistence in corridor choice is the main goal of this proposal.

This paper studies corridor allocation for East African migration flows toward Europe through the two dominant irregular-crossing corridors: the Eastern Mediterranean route (via Turkey) and the Central Mediterranean route (via Libya and Tunisia). The empirical focus is on the composition of flows across destination-route alternatives, conditional on observed migration activity. The key identifying variation comes from three major conflict escalations during the sample period: the Tigray War in northern Ethiopia (November 2020), the Gaza conflict beginning in October 2023, and the Sudan civil war (April 2023). These shocks affect distinct segments of the two corridors, providing within-corridor variation in transit risk.

The contribution that the proposal aims to make is twofold. First, constructing a continuous, network-weighted conflict traversal cost for migration routes using geocoded UCDP conflict events overlaid on the road network, replacing coarser conflict measures common in the conflict-and-migration literature such as (Melander and Öberg, 2007) where there is an actual area of effect and it can be appreciated which roads are affected. This measure captures the intensity and spatial distribution of violence along transit corridors, not merely its presence. Second, the paper documents persistence in corridor-level migration shares and estimates how conflict shocks redistribute flows, using a staged empirical design: a dynamic reduced-form model with event-time diagnostics as the primary analysis, and a structural mixed-logit extension that, once route cost are available (not available yet due to not enough computational resources), can recover heterogeneous conflict sensitivity. Migrants who continue to choose more conflict-exposed corridors are interpreted as less conflict-averse in relative terms, while the lagged-share term captures corridor lock-in and persistence.

The main missing piece is the route-cost panel: the script exists, but the full panel has not yet been materialised because the least-cost-path step is too computationally-intensive. The remaining work is therefore focused on computing route costs on a higher-memory machine, adding route-level controls, and then running the mixed-logit stage.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 defines the estimand and scope. Section 4 describes the data architecture and measurement strategy. Section 5 presents the econometric design. Section 6 reports preliminary results. Section 7 discusses the main limitations and the checks used to address them. Section 8 outlines the remaining work plan.

2 Related Literature

This proposal aims to contribute to three strands of the migration literature: the economics of forced migration, the structural modelling of destination choice, and the literature on migration routes and transit risk.

2.1 Forced Migration and Conflict

A large body of work documents the relationship between conflict and migration flows. Melander and Öberg (2007) establish that the threat of violence, irrespective of its intensity, is a significant predictor of forced displacement. Miguel, Satyanath, and Sergenti (2004) demonstrate the causal link between economic shocks and civil conflict.

The forced migration literature has increasingly emphasised that displacement decisions involve not only whether to leave but also where to go and how to get there (Ibáñez and Vélez, 2008; Ruiz and Vargas-Silva, 2013). Dustmann et al. (2017) provide a broad economic framework for thinking about refugee migration, policy, and destination choice under displacement.

This paper proposes an extension to this work by constructing a continuous measure of transit risk and estimating the degree to which established corridors exhibit persistence in the face of changing conditions.

2.2 Structural Models of Migration Choice

The structural approach to migration destination choice builds on the discrete-choice framework of McFadden (1974), extended to aggregate settings by Berry (1994) and Berry, Levinsohn, and Pakes (1995). Bertoli and Fernández-Huertas Moraga (2013) adapt this framework to international migration, introducing multilateral resistance to migration as the analogue of multilateral resistance to trade in the gravity framework. Their approach models destination shares as a function of bilateral costs and destination attractiveness, allowing for correlation across alternatives through nested and mixed-logit specifications.

Beine, Docquier, and Özden (2011) use aggregate bilateral migration data to estimate how diaspora networks, migration policies, and economic conditions shape destination choices, highlighting the strong role of existing diasporas in bilateral migration flows. Beine and Parsons (2015) examine climatic and disaster-related determinants of international migration within a bilateral gravity-style framework. Ortega and Peri (2013) estimate the effects of immigration policy and income differences on bilateral migration flows using a panel gravity model. Grogger and Hanson (2011) develop a model of migrant selection and sorting across destinations in which returns to skill differ across countries.

This paper differs from the standard bilateral migration-choice literature in two respects. First, the choice set includes destination-route pairs rather than destinations alone, because the transit corridor is a decision margin for irregular migrants. Second, the key variation comes from conflict exposure along transit corridors, not from destination-level wage gaps or policy differences.

2.3 Migration Routes and Transit Risk

An emerging literature studies the role of transit conditions in shaping migration flows. León and Miguel (2016) study risky transportation choices more generally, highlighting behavioural responses to mortality risk. Friebe and Guriev (2006) model the market for smuggling services and show that transit risk affects both the volume and composition of migration flows. de Haas (2021) argues that migration systems exhibit internal dynamics—path dependence in corridor use driven by network effects, infrastructure, and smuggling networks—that generate persistence even when conditions change.

The role of border enforcement in redirecting flows has been documented extensively. The EU–Turkey agreement of March 2016 dramatically reduced Eastern Mediterranean crossings, while the Italy–Libya Memorandum of Understanding of February 2017 affected Central Mediterranean flows (Frontex, 2023). McKenzie and Rapoport (2010) document self-selection patterns in migration that are shaped by networks and migration costs in the Mexico–US context. Munshi (2020) provides a helpful understanding on how networks sustain migration corridors over time.

This proposal aims to contribute to the route-choice literature by constructing a spatially explicit, time-varying measure of corridor risk and estimating the persistence-reallocation trade-off directly. The inclusion of the Gaza conflict (October 2023) as a shock affecting the Eastern Mediterranean corridor complements the Sudan civil war shock on the Central Mediterranean side, providing identification from contemporaneous shocks hitting different corridors.

3 Scope and Estimand

3.1 Main Estimand

The main estimand is the conditional association between route conflict exposure and destination-route shares:

$$\Delta s_{ojt} \text{ associated with } \Delta \text{RouteCost}_{ort},$$

where $j = (d, r)$ denotes a destination-route alternative. This estimand has for now an associational interpretation and not a causal one in the design shown as the route costs cannot be computed due to being too computationally intensive. Section 7 discusses the main threats and the checks used to address them.

In the current model, the mixed-logit stage introduces heterogeneity only through the route-cost coefficient. Persistence still enters as a common lagged-share term. I retain that mixed-logit structure because route choice can still reflect two forces, conflict sensitivity and persistence, but the implemented random-coefficient stage identifies heterogeneity only in the first of them.

3.2 Interpretation Boundary

The baseline design identifies route allocation conditional on observed migration flows, not a complete behavioural model of the leave/stay migration margin. Origin-conflict effects on total

migration volume are treated as a separate complementary equation.

3.3 Geographic and Temporal Scope

- **Origins:** Ethiopia, Eritrea, Sudan, Somalia, South Sudan, Kenya, Djibouti.
- **Destinations:** EU member states and selected European destinations observed in Eurostat asylum application data (31 countries).
- **Routes:** Eastern Mediterranean (via Turkey, entry point near Izmir) and Central Mediterranean (via Libya/Tunisia, entry points near Tripoli and Sfax).
- **Time:** Monthly panel from October 2011 to March 2024 ($T \approx 150$ months per corridor cell).

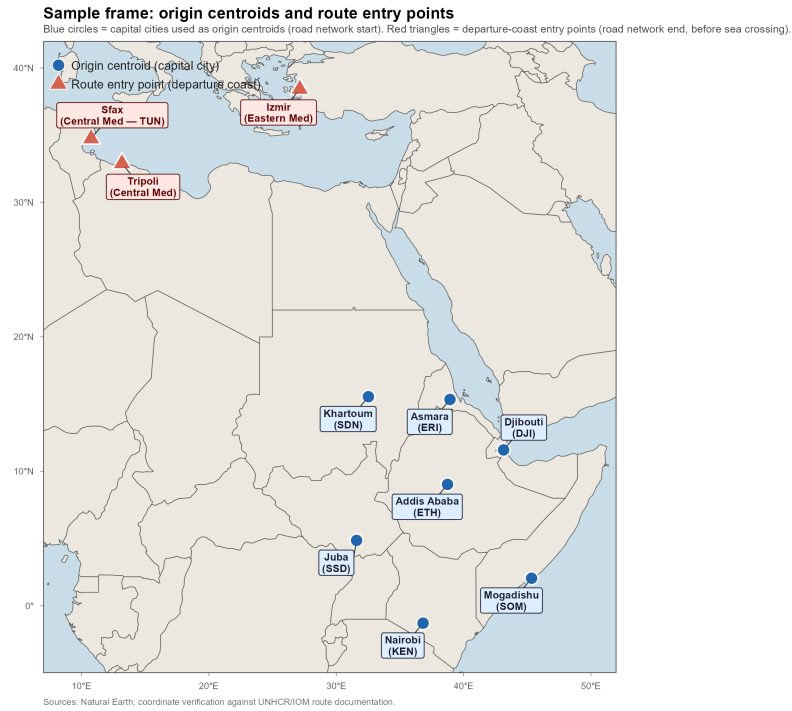


Figure 1: Sample frame: origin centroids and route entry points. Blue circles mark the seven East African origin countries (capital cities used as road network start nodes). Red triangles mark the Eastern Mediterranean (Izmir) and Central Mediterranean (Tripoli, Sfax) departure-coast entry points used in the route-cost computation.

4 Data Architecture and Measurement

4.1 Data Sources

Table 1: Core data sources

Source	Frequency	Unit	Role
Eurostat asylum applications	Monthly	Origin-destination applications	OD margin for IPFP and outcome-share construction
Frontex IBC detections	Monthly	Origin-route detections	Route margin for IPFP and observed corridor allocation by nationality
UCDP GED v24.1	Event-level	Geocoded conflict events (location, fatalities, type)	Route-conflict cost construction
UCDP/PRIO Armed Conflict	Annual	Country-year	Origin conflict controls
UNHCR Persons of Concern	Annual	Bilateral stocks	External validation benchmark
Abel–Cohen bilateral estimates (Abel and Cohen, 2019)	5-year snapshots	Bilateral rates	Prior weights for IPFP initialisation

4.2 Latent Origin-Destination-Route Reconstruction

Bilateral destination-route flows are not directly observed. The asylum application data (Eurostat) record origin-destination flows, while the border detection data (Frontex) record route totals and, in many periods, nationality-route cells without destination disaggregation. To construct the estimation panel, each period’s latent flow matrix Z_{odrt} is inferred by iterative proportional fitting (IPFP), imposing:

$$\sum_r Z_{odrt} = F_{odt} \quad (\text{Eurostat OD totals}), \quad (1)$$

$$\sum_{o,d} Z_{odrt} = C_{rt} \quad (\text{Frontex route totals}). \quad (2)$$

The baseline convention treats OD totals as authoritative and rescales route totals to match aggregate OD volume before balancing. The initialisation uses Abel–Cohen bilateral migration estimates (Abel and Cohen, 2019) as prior weights, with a minimum positive floor to avoid structural-zero infeasibility.

Lag-correlation diagnostics between monthly OD totals and monthly route detections reveal that the peak correlation occurs at a one-month lead ($r = 0.787$ when OD applications lead route detections by one month), with a contemporaneous correlation of $r = 0.719$. This timing pattern is consistent with the administrative lag between border crossing and asylum application

filing.

4.3 Generated-Regressand Problem

Because the destination-route dependent variable Z_{odrt} is the output of an IPFP procedure rather than a direct observation, any logit estimated on this panel is conditional on the IPFP reconstruction. In the current build, this issue is handled mainly by keeping the observed Frontex route margin visible, reporting IPFP diagnostics, and treating destination-route estimates as descriptive. A full bootstrap over the IPFP-plus-estimation pipeline remains a planned extension.

4.4 Observed Corridor Allocation by Origin

Although destination-route flows are latent, the corridor margin by nationality is directly observed in the Frontex IBC data. Figure 2 plots observed East African nationality-to-corridor allocations for the post-Sudan-war window (April 2023–March 2024) for the highest-volume origins in the sample. This figure is useful as it shows the empirical object that the mixed logit is trying to rationalise: why some origins load more heavily onto the Central Mediterranean corridor while others remain on the Eastern Mediterranean route.

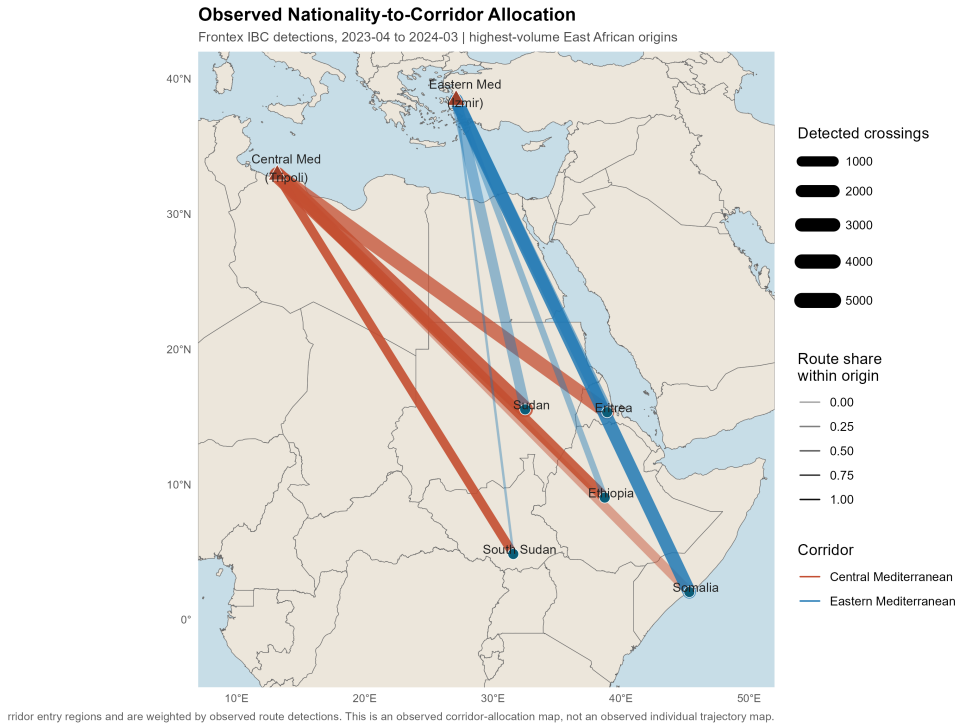


Figure 2: Observed nationality-to-corridor allocation, April 2023–March 2024. Lines connect origin capitals to Frontex corridor entry regions and are weighted by total observed route detections for the highest-volume East African origins in the post-Sudan-war window.

We can also visualize computing the shortest path and with the data on the roads how does the route look like for different countries previous to the inclusion of the route cost that come

with conflict, these are not observed migrant trajectories they are but shortest land paths to the corridor entry points with the sea crossing omitted.

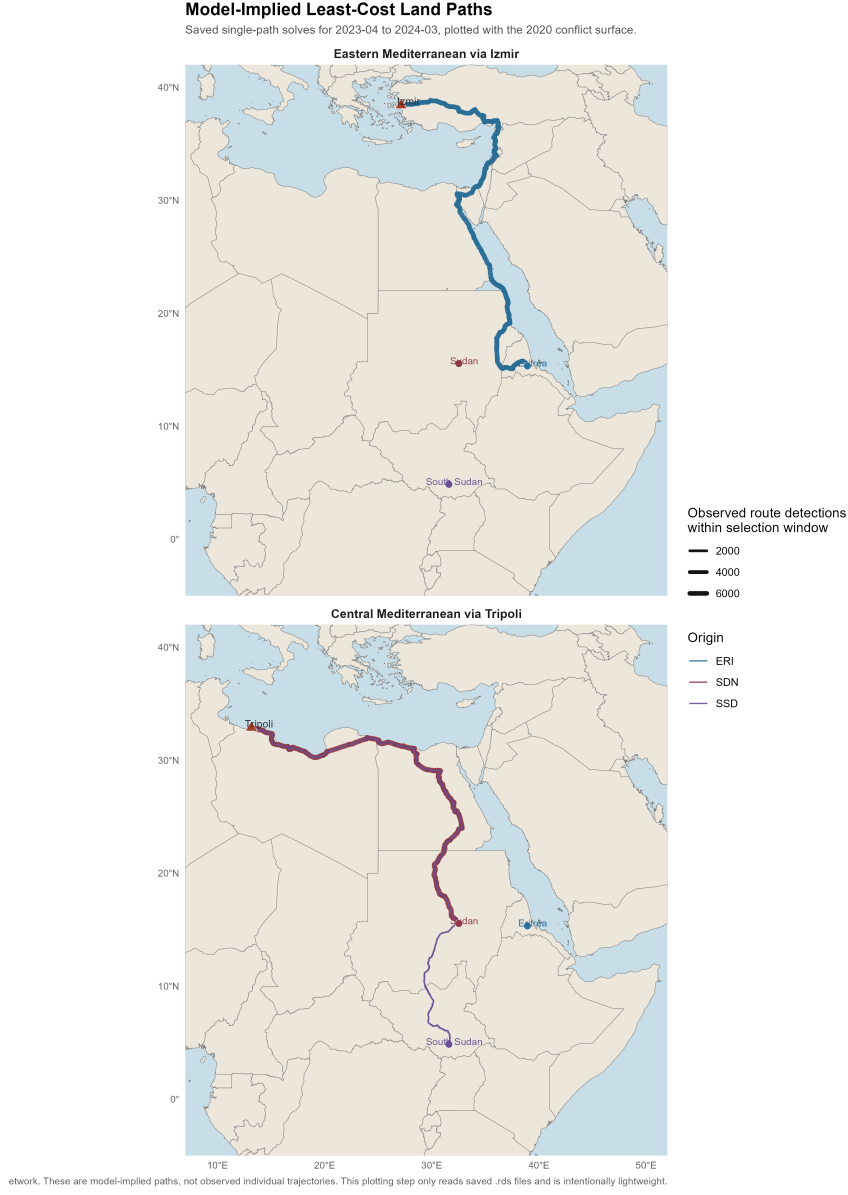


Figure 3: Each line is a conflict-weighted shortest land path from an origin capital to a corridor departure point, computed separately and then combined into one figure. The Eastern Mediterranean panel shows the saved Eritrea route to Izmir. The Central Mediterranean panel shows the saved Sudan and South Sudan routes to Tripoli. The figure is illustrative and does not represent observed individual trajectories.

4.5 Conflict-Weighted Route Cost

For each year t , route conflict exposure is constructed by:

- rasterising UCDP GED fatalities over the corridor region (7°E – 52°E , 5°S – 42°N) using a Gaussian kernel (1° bandwidth, 0.1° grid), normalised by the corridor-wide 99th percentile over 2010–2024

- (ii) extracting conflict intensity at road network edge centroids
- (iii) assigning edge costs:

$$c_{et} = d_e (1 + \lambda \cdot \text{ConflictDensity}_{et}), \quad (3)$$

where d_e is edge length and λ is a scaling parameter calibrated through sensitivity grids

- (iv) computing minimum-cost paths from origin centroids to route entry points using Dijkstra's algorithm on the road network.

This yields RouteCost_{ort} at annual frequency, merged to the monthly panel by calendar year. The corridor bounding box encompasses the Israel–Palestine region, so the October 2023 Gaza escalation is captured in the conflict rasters for 2023 and 2024, directly affecting Eastern Mediterranean route costs for origins whose least-cost paths traverse the eastern corridor.

Computational feasibility is the main problem at this and in the whole proposal. The Constructing a raster and assigning the costs of each conflict in it is not doable with my current resources and neither with some of my colleagues devices. I therefore treat route-cost construction as a staged task: first compute the full panel on a higher-memory machine, then merge it into the estimation dataset, and only then run the mixed-logit stage. If the full 2010–2024 panel remains infeasible, the fallback is a reduced event-year panel centred on 2020, 2023, and 2024.

Figure 4 illustrates the spatial distribution of conflict intensity for six years spanning the main identification-relevant shocks.

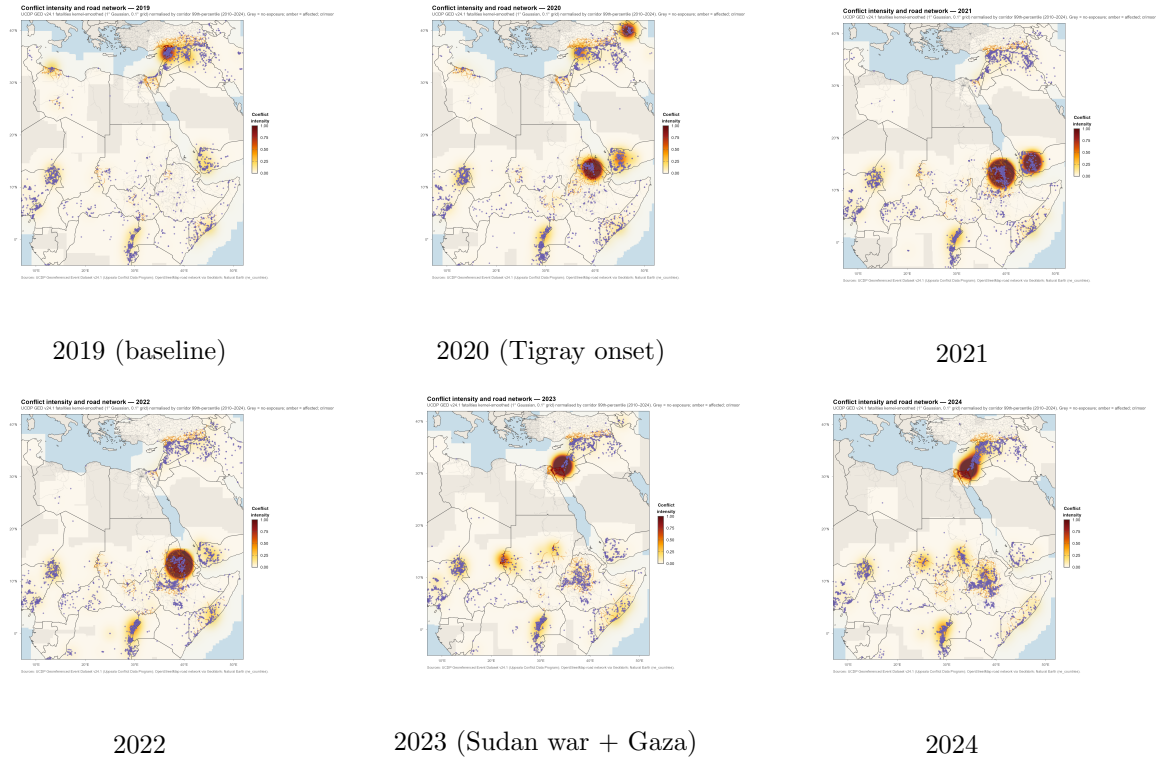


Figure 4: Conflict intensity and road network sequence, 2019–2024. The 2020 panel shows the Tigray War escalation in northern Ethiopia. Blue points mark individual GED events.

5 Econometric Design

The empirical strategy proceeds in three stages, with the reduced-form analysis and event-time diagnostics as the primary components.

5.1 Stage 1: Dynamic Conditional Logit (Baseline)

The baseline specification estimates a conditional aggregate logit on market shares:

$$s_{ojt} = \frac{\exp(V_{ojt})}{\sum_{j'} \exp(V_{oj't})}, \quad (4)$$

where the deterministic component of utility is:

$$\begin{aligned} V_{ojt} = & \alpha_j + \beta_1 \text{RouteCost}_{ort} \\ & + \beta_2 (\text{OriginConflict}_{ot} \times \mathbf{1}[r = \text{Central}]) \\ & + \rho \ln(s_{oj,t-1} + \kappa) + \tau_t. \end{aligned} \quad (5)$$

Origin conflict is interacted with route type because pure origin-time terms are common across alternatives within market (o, t) and are not identified in a conditional share framework. Additionally, common time fixed effects τ_t absorb aggregate shocks, richer alternative-time structures are explored as robustness checks.

The inclusion of a lagged dependent variable alongside alternative-specific fixed effects introduces Nickell-type bias. With $T \approx 150$ monthly periods per corridor cell, Nickell bias alone is small (≈ 0.007), so the large $\hat{\rho}$ reported below cannot be attributed to it, it is driven by persistent unobservables that alternative-specific fixed effects do not absorb. Sensitivity checks are implemented via dynamic-panel estimators (Anderson and Hsiao, 1982; Arellano and Bond, 1991).

The current implementation estimates this baseline with alternative fixed effects, the lagged share, and the origin-conflict interaction. Route cost enters only if it is available, which as per not enough computational resources, it is not, so the reported estimates are from the reduced model without β_1 . Border-policy and transit-environment controls are important confounders, but they are not yet implemented and should therefore be read as next-step extensions rather than current regressors.

Further expansion: Persistence measurement out of the utility function entirely and estimate it on the observed Frontex nationality-route panel, which does not depend on the IPFP reconstruction.

5.2 Stage 2: Event-Time Diagnostics

Event-time analysis is constructed around three conflict escalations:

- (i) the Tigray War onset, November 2020;
- (ii) the Sudan civil war onset, April 2023;
- (iii) the Gaza conflict escalation, October 2023.

For each event, route shares are plotted in a 24-month pre-event and as possible months, post-event window, with pre-post mean differences computed over 12-month symmetric windows. These diagnostics serve as transparent timing evidence and complement the model-based estimates.

Because route cost is annual and policy controls are not yet merged, the event-time evidence should be read as timing evidence rather than as a formal identification design. A stacked difference-in-differences specification is a possible next extension once the route-cost and policy series are in place.

5.3 Stage 3: Structural Mixed-Logit Extension

Conditional indirect utility for latent type g :

$$\begin{aligned}
 U_{gojt} = & \alpha_j + \beta_{\text{lag}} \ln(s_{oj,t-1} + \kappa) \\
 & + \beta_{\text{conf}} (\text{OriginConflict}_{ot} \times \mathbf{1}[r = \text{Central}]) \\
 & + (\bar{\theta} + \sigma_{\theta} \nu_g) \text{RouteCost}_{ort} + \varepsilon_{gojt}.
 \end{aligned} \tag{6}$$

Shares integrate over the type distribution:

$$s_{ojt} = \int \frac{\exp(U_{gojt})}{\sum_{j'} \exp(U_{goj't})} dF(g).$$

The current mixed extension is narrower than the conceptual motivation in the introduction. In the present implementation, heterogeneity enters only through the route-cost coefficient, the lag term remains common. The mixed logit therefore estimates heterogeneity in conflict sensitivity conditional on common persistence, not heterogeneity in persistence itself. Since route cost is not available for now the estimate, falls back to the MNL and skips the mixed stage.

To avoid mechanical scale confounding, the route-cost scaling parameter λ in Equation (3) is fixed through calibration grids while θ_g is estimated. Standard errors are obtained from the numerical Hessian.

Three limitations should be noted. First, with only two routes per destination, the choice set is effectively binary within (o, d, t) , The mixed logit is therefore best read as a two-type heterogeneous corridor choice, capturing that some migrants remain on high-conflict corridors even when others switch, not as a formal recovery of a full behavioural distribution. Second, the route-cost variable measures aims to measure conflict-weighted traversal cost, not mortality

risk directly, interpreting θ_g as capturing differential conflict-sensitivity requires a maintained mapping between traversal cost and physical risk. For that reason, I interpret θ_g as relative conflict-sensitivity, and only conditionally as risk aversion.

6 Preliminary Results

This section reports the preliminary results from the proposal. All estimates should be interpreted cautiously, as the route-cost variable has not yet been integrated into the estimation panel. The baseline therefore captures persistence and origin-conflict interactions without the paper’s central structural regressor. These estimates are used to calibrate the persistence problem before the full mixed-logit route-cost stage is run.

6.1 Conditional Logit Baseline

Table 2 reports the main parameter estimates from the conditional aggregate logit. The estimation includes 43 parameters: 41 alternative-specific fixed effects (for 21 EU destinations $\times$ 2 routes, with one normalised to zero), a persistence coefficient, and an origin-conflict $\times$ Central Mediterranean interaction.

Table 2: Conditional aggregate logit baseline estimates

Parameter	Standardised	Raw scale
Persistence ($\hat{\rho}$): lagged share	2.796	1.013
Origin conflict $\times$ Central Med ($\hat{\beta}_2$)	−0.00038	−0.00049
Log-likelihood	−1,480,727	
Weighted RMSE	0.0016	
Weighted MAE	0.0010	
Mean market KL divergence	0.0018	
Number of parameters	43	
Fit type	MNL (no route cost)	

Notes: Standardised coefficients are in units of the within-sample standard deviation of each covariate. Raw-scale coefficients are rescaled by covariate standard deviation. The estimation uses weighted log-likelihood with market size as weights.

Three features can be highlighted. First, the persistence coefficient is very large ($\hat{\rho} = 1.01$ in raw units), indicating that lagged shares are the dominant predictor of current shares. Second, the origin-conflict interaction is economically and statistically negligible (−0.00049), suggesting that without route-cost variation, there is no measurable conflict-driven reallocation in the conditional model. Third, in-sample fit is extremely high (weighted RMSE = 0.0016, mean market KL divergence = 0.0018), which is driven by the rich alternative fixed effects and persistence term rather than structural variation in conflict exposure.

6.2 Dynamic-Panel Sensitivity

The magnitude of the persistence parameter is sensitive to estimator choice. Table 3 compares the baseline MNL estimate with two dynamic-panel diagnostics.

Table 3: Dynamic-panel diagnostics for the persistence parameter

Estimator	$\hat{\rho}$	Std. error	Note
Conditional logit (MNL)	1.013	—	Baseline, potential Nickell bias
Anderson–Hsiao (2SLS)	—	—	First-stage fit failed
Arellano–Bond (two-step)	0.098	0.051	GMM, instruments: y_{t-2}, y_{t-3}

Notes: The Anderson–Hsiao estimator instruments the first-differenced lagged dependent variable with y_{t-2} , the first-stage regression failed in the current build (30,366 observations). The Arellano–Bond estimator uses y_{t-2} and y_{t-3} as instruments.

The Arellano–Bond estimate ($\hat{\rho} = 0.098$, $SE = 0.051$) is an order of magnitude smaller than the conditional logit estimate, which on its own indicates that the MNL lag is absorbing persistent corridor-level unobservables that the alternative fixed effects do not soak up. This gap is interpreted as a specification verdict rather than a sensitivity finding: the conditional-logit persistence is an upper bound on the combination “true state dependence + correlated unobservables” and is not reported as a point estimate. We can interpret Table 3 is therefore limited but clear: the MNL lag is too large to be treated as a structural persistence estimate, and a cleaner route-margin persistence step is still needed before the mixed-logit stage can be interpreted confidently.

6.3 Event-Time Evidence

Table 4 reports the two conflict onsets for which the current sample contains a full 12-month post-event window. The figure below additionally shows the Gaza escalation, but that event still has a short post period in the current data build.

Table 4: Route-share reallocation around major conflict onsets

Event	Route	Pre-event mean	Post-event mean	Δ (pp)
Tigray War (Nov 2020)	Central Med	0.703	0.715	+1.2
	Eastern Med	0.297	0.285	−1.2
Sudan War (Apr 2023)	Central Med	0.489	0.647	+15.8
	Eastern Med	0.511	0.353	−15.8

Notes: Pre-event and post-event means computed over symmetric 12-month windows. Shares are aggregated across all destinations within each origin-route-month cell. The Gaza conflict (October 2023) is plotted in the event-time figure, but it is omitted from the table because the post-event window is still short.

The table shows the strongest evidence for the Sudan war: Central Mediterranean shares increase from approximately 49% to 65% in the year following onset, with a mirror decline in Eastern Mediterranean shares. This is consistent with Sudan’s role as a major transit country for the Central Mediterranean corridor (via Libya), where the civil war simultaneously disrupted

transit infrastructure and generated new outflows, which could also be complementary to the Gaza conflict. The Tigray onset, by contrast, produced a modest 1.2 percentage point shift, consistent with northern Ethiopian conflict primarily affecting origin push factors rather than corridor transit conditions.

6.4 Route-Share Persistence

Figure 5 displays the route-share sequence for the full 2019–2024 window, both in aggregate and by origin country.

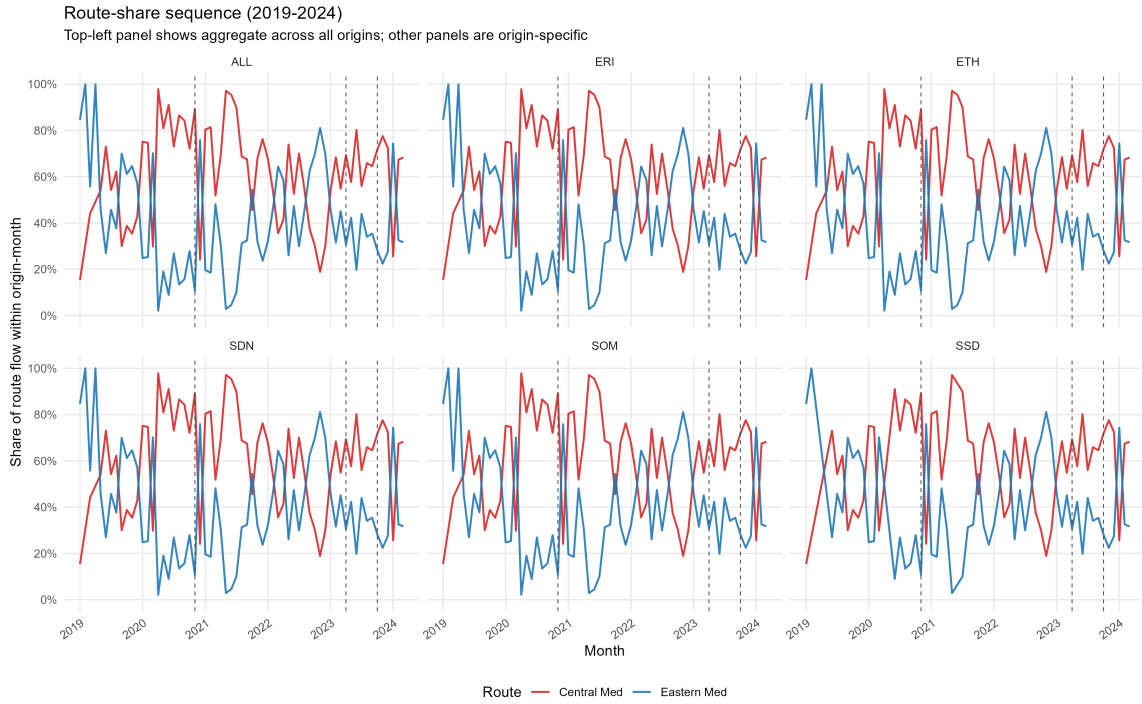


Figure 5: Route-share sequence, 2019–2024.

The origin-specific panel show substantial persistence in corridor allocation, with visible reallocations during the Sudan war period. The persistence pattern is consistent across most origins, though the magnitude of reallocation varies substantially, we can see that it is most pronounced for Sudanese and Eritrean flows.

6.5 Data Alignment Diagnostics

The timing relationship between the two data systems (Eurostat applications and Frontex detections) is documented in Table 5. The peak cross-correlation at a one-month lead supports the interpretation that border detections precede asylum application recording, validating the IPFP assumption that contemporaneous margins are approximately aligned.

Table 5: Lag-correlation between OD applications (Eurostat) and route detections (Frontex)

Lag (months)	Correlation
−3	0.664
−2	0.754
−1	0.787
0	0.719
+1	0.591
+2	0.456
+3	0.319

7 Limitations

- **Persistence is not yet cleanly identified.** Lagged shares may capture true corridor lock-in, but they may also pick up persistent omitted route traits. The current gap between the conditional logit and Arellano–Bond persistence estimates shows that this issue is important. The practical check is to compare those diagnostics and then estimate persistence directly on the observed Frontex nationality-route panel.
- **The route shares are reconstructed with IPFP.** That means results depend on priors, support restrictions, and margin scaling. The validation response is to rerun the reconstruction under a small set of alternative priors and compare inferred nationality-route shares to observed Frontex cells wherever those cells are available.
- **Policy, enforcement, and smuggling conditions can also move routes.** This is the main confounding risk in the reduced form. The response is to add a small set of route-specific policy and enforcement controls, use exclusion windows around major policy breaks, and keep the interpretation of the current estimates cautious until those controls are merged.
- **Some checks are still pending because route cost are not yet possible to compute.** The main remaining checks are alternative IPFP priors, alternative lag choices, reduced event-year route-cost panels if the full panel remains too heavy, and the mixed-logit extension once route cost becomes available.

8 Roadmap

1. Estimate route persistence directly on the observed Frontex nationality-route panel.
2. Compute the route-cost panel on a higher-memory computer, or use a smaller event-year panel if the full panel remains infeasible.
3. Add a compact set of policy, enforcement, and transit controls to the reduced-form model.

4. Run a focused robustness set: alternative IPFP, alternative lag choices, and key sample restrictions.
5. Finalise the main figures for the draft, including the route allocation maps, event-time plots, and model-implied path maps where the road graph is connected.
6. Estimate the mixed-logit extension once route cost is available and then complete the dissertation chapter and replication package.

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